

0.1 Cauchy's Integral Theorem, Cauchy's Integral Formula, Morera's Theorem, and Liouville's Theorem

The geometric rigidity of holomorphic functions allows for a profound connection between complex contour integration and the values of a function entirely strictly within a domain.

[Cauchy's Integral Theorem] Let $\Omega \subset \mathbb{C}$ be a simply connected open set, and let $f : \Omega \rightarrow \mathbb{C}$ be a holomorphic function. If γ is a piecewise smooth, simple closed contour contained entirely within Ω , then:

$$\oint_{\gamma} f(z) dz = 0$$

We present the classical proof under the assumption that $f'(z)$ is continuous, which permits the use of Green's Theorem (Goursat later demonstrated that the theorem holds without assuming continuous partial derivatives). Let $f(z) = u(x, y) + iv(x, y)$ and let $dz = dx + idy$. We separate the contour integral into real and imaginary line integrals:

$$\oint_{\gamma} f(z) dz = \oint_{\gamma} (u + iv)(dx + idy) = \oint_{\gamma} (u dx - v dy) + i \oint_{\gamma} (v dx + u dy)$$

Because Ω is simply connected and γ is a simple closed contour, γ bounds a simply connected region $D \subset \Omega$. We apply Green's Theorem to both real line integrals:

$$\begin{aligned} \oint_{\gamma} (u dx - v dy) &= \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy \\ \oint_{\gamma} (v dx + u dy) &= \iint_D \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy \end{aligned}$$

Because f is holomorphic on Ω , it satisfies the Cauchy-Riemann equations: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. Substituting these into the double integrals yields integrands of identically zero:

$$\iint_D (0) dx dy + i \iint_D (0) dx dy = 0$$

Thus, $\oint_{\gamma} f(z) dz = 0$.

[Cauchy's Integral Formula] Let f be holomorphic on a simply connected open set Ω , and let γ be a simple closed contour in Ω oriented counterclockwise. For any point z_0 strictly in the interior of γ :

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz$$

Because Ω is open and z_0 lies in the interior of γ , there exists an $\epsilon > 0$ such that the closed disk $\overline{D_{\epsilon}(z_0)} = \{z \in \mathbb{C} \mid |z - z_0| \leq \epsilon\}$ is entirely contained within the interior of γ . Let C_{ϵ} denote the boundary circle of this disk, parametrized by $z(\theta) = z_0 + \epsilon e^{i\theta}$ for $0 \leq \theta \leq 2\pi$, oriented counterclockwise.

The function $g(z) = \frac{f(z)}{z-z_0}$ is holomorphic everywhere in Ω except at z_0 . By the principle of contour deformation (a corollary of Cauchy's Integral Theorem on multiply connected domains), the integral over γ equals the integral over C_ϵ :

$$\oint_{\gamma} \frac{f(z)}{z-z_0} dz = \oint_{C_\epsilon} \frac{f(z)}{z-z_0} dz$$

We rewrite the integrand by adding and subtracting $f(z_0)$:

$$\oint_{C_\epsilon} \frac{f(z)}{z-z_0} dz = \oint_{C_\epsilon} \frac{f(z_0)}{z-z_0} dz + \oint_{C_\epsilon} \frac{f(z) - f(z_0)}{z-z_0} dz$$

We evaluate the first integral by direct parametrization ($dz = i\epsilon e^{i\theta} d\theta$):

$$\oint_{C_\epsilon} \frac{f(z_0)}{z-z_0} dz = f(z_0) \int_0^{2\pi} \frac{i\epsilon e^{i\theta}}{\epsilon e^{i\theta}} d\theta = f(z_0) \int_0^{2\pi} i d\theta = 2\pi i f(z_0)$$

For the second integral, because f is holomorphic, it is continuous at z_0 . For any $\eta > 0$, we can choose ϵ sufficiently small such that $|f(z) - f(z_0)| < \eta$ for all $z \in C_\epsilon$. We bound the second integral using the standard estimation lemma (ML-inequality):

$$\left| \oint_{C_\epsilon} \frac{f(z) - f(z_0)}{z-z_0} dz \right| \leq \max_{z \in C_\epsilon} \left| \frac{f(z) - f(z_0)}{z-z_0} \right| \cdot \text{length}(C_\epsilon) < \frac{\eta}{\epsilon} (2\pi\epsilon) = 2\pi\eta$$

Because the original integral over γ is independent of ϵ , and η can be made arbitrarily small by shrinking ϵ , the second integral must be exactly 0. Therefore:

$$\oint_{\gamma} \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0) + 0$$

Dividing by $2\pi i$ yields the result.

[Liouville's Theorem] If $f : C \rightarrow \mathbb{C}$ is an entire function (holomorphic on all of C) and is bounded, then f is constant.

Because f is bounded, there exists a real number $M > 0$ such that $|f(z)| \leq M$ for all $z \in C$. Fix an arbitrary point $z_0 \in C$. By differentiating Cauchy's Integral Formula with respect to z_0 , we obtain an integral representation for the derivative:

$$f'(z_0) = \frac{1}{2\pi i} \oint_{C_R} \frac{f(z)}{(z-z_0)^2} dz$$

where C_R is a circle of radius $R > 0$ centered at z_0 . We apply the estimation lemma to bound the magnitude of the derivative:

$$|f'(z_0)| \leq \frac{1}{2\pi} \max_{z \in C_R} \left| \frac{f(z)}{(z-z_0)^2} \right| \cdot \text{length}(C_R)$$

Since $|f(z)| \leq M$ and $|z - z_0| = R$ on the contour C_R , this yields:

$$|f'(z_0)| \leq \frac{1}{2\pi} \frac{M}{R^2} (2\pi R) = \frac{M}{R}$$

Because f is entire, this inequality holds for any arbitrarily large radius R . Taking the limit as $R \rightarrow \infty$, we find $|f'(z_0)| \leq 0$, forcing $f'(z_0) = 0$. Since z_0 was chosen arbitrarily, $f'(z) = 0$ for all $z \in C$. A function with a strictly zero derivative on a connected domain is constant.

[Morera's Theorem] Let $\Omega \subset C$ be an open set. If $f : \Omega \rightarrow C$ is continuous and satisfies $\oint_{\gamma} f(z) dz = 0$ for every closed piecewise smooth curve γ in Ω , then f is holomorphic on Ω .

Because the integral of f over any closed loop is zero, the integral between any two points is path-independent. Fix $z_0 \in \Omega$ and define a function $F : \Omega \rightarrow C$ by:

$$F(z) = \int_{z_0}^z f(w) dw$$

This function is well-defined due to path independence. We compute the derivative of F directly from the definition:

$$\lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \int_z^{z+h} f(w) dw$$

Because f is continuous at z , we can write $f(w) = f(z) + (f(w) - f(z))$. Substituting this into the integral:

$$\frac{1}{h} \int_z^{z+h} [f(z) + (f(w) - f(z))] dw = f(z) + \frac{1}{h} \int_z^{z+h} (f(w) - f(z)) dw$$

As $h \rightarrow 0$, w approaches z . By the continuity of f , for any $\epsilon > 0$, there exists $\delta > 0$ such that $|w - z| < \delta \implies |f(w) - f(z)| < \epsilon$. For $|h| < \delta$, the error term is bounded by:

$$\left| \frac{1}{h} \int_z^{z+h} (f(w) - f(z)) dw \right| \leq \frac{1}{|h|} \cdot \epsilon \cdot |h| = \epsilon$$

Thus, the limit evaluates exactly to $f(z)$, proving that $F'(z) = f(z)$. Therefore, F is a holomorphic function on Ω . A profound corollary of Cauchy's Integral Formula is that the derivative of a holomorphic function is itself holomorphic (analytic functions are infinitely differentiable). Since $f = F'$, f must also be holomorphic.